Fundamental Theorem of Calculus:

$$\int_{a}^{b} f(x)dx = F(b) - F(a)$$

1) Evaluate the definite integral $\int_0^2 (3x) \cdot dx$

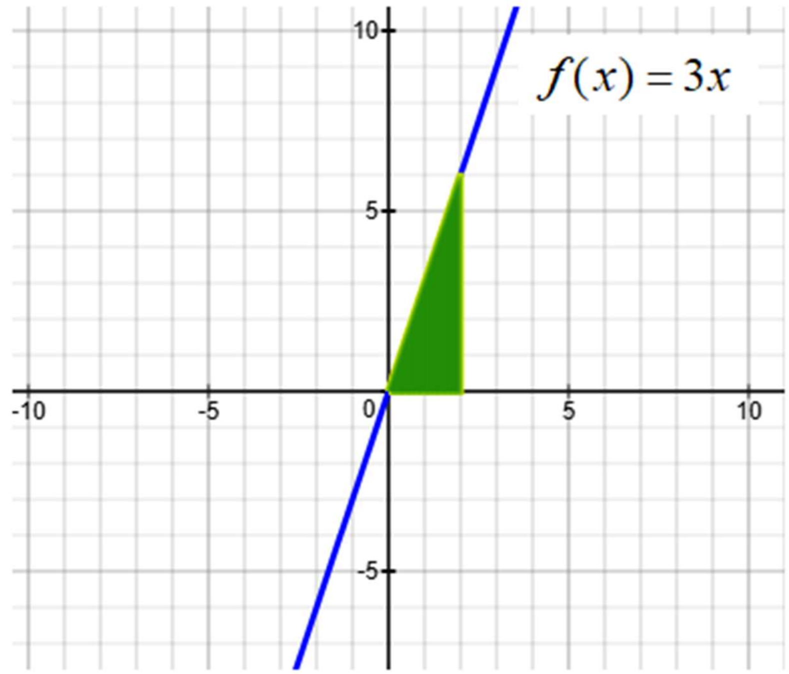
a) Antiderivate $F(x) = \int 3x \cdot dx = \underline{\quad ? \quad}$

b) $F(2) = \underline{\quad ? \quad}$

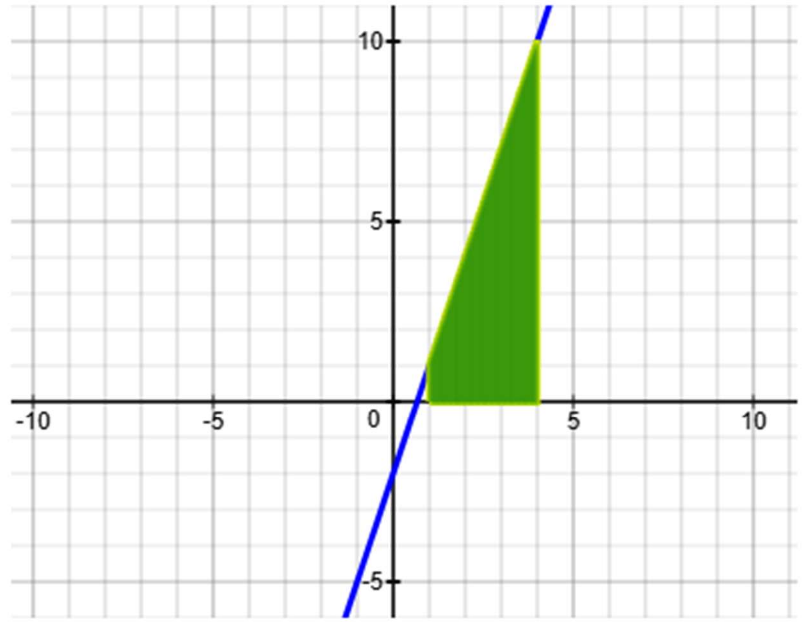
c) $F(0) = \underline{\quad ? \quad}$

d) $\int_0^2 (3x) \cdot dx = F(2) - F(0) = \underline{\quad ? \quad}$

e) Area of shaded region $= \underline{\quad ? \quad}$



2) Evaluate the definite integral $\int_{1}^{4}(3x-2)\cdot dx$



a) Antiderivate $F(x)=\int(3x-2)\cdot dx=$ ___?___

b) $F(4)=$ ___?___

c) $F(1)=$ ___?___

d) $\int_{1}^{4}(3x-2)\cdot dx=F(4)-F(1)=$ ___?___

e) Area of shaded region $=$ ___?___

Note: Area above x-axis is positive.

Area below x-axis is negative.

3) Evaluate the definite integral $\int_{-1}^{1}\left(x^2-1\right)\cdot dx$

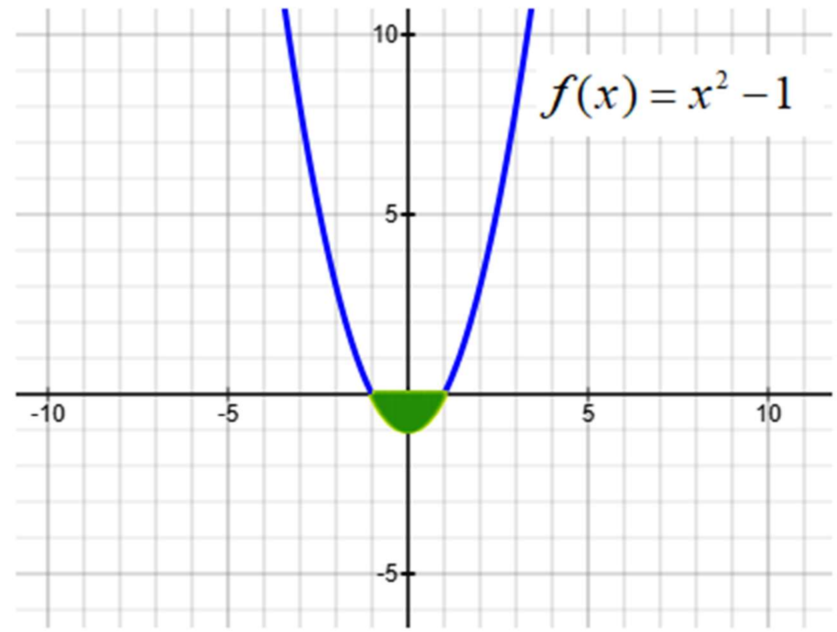
a) Antiderivate $F(x)=\int\left(x^2-1\right)\cdot dx=$ ___?___

b) $F(1)=$ ___?___

c) $F(-1)=$ ___?___

d) $\int_{-1}^{1}\left(x^2-1\right)\cdot dx=F(1)-F(-1)=$ ___?___

e) Area of shaded region $=$ ___?___



Note: Area above x-axis is positive.

Area below x-axis is negative.

4) Evaluate the definite integral $\int_0^1 (2x-1)^2 \cdot dx$

a) Antiderivate $F(x) = \int (2x-1)^2 \cdot dx = \underline{\quad ? \quad}$

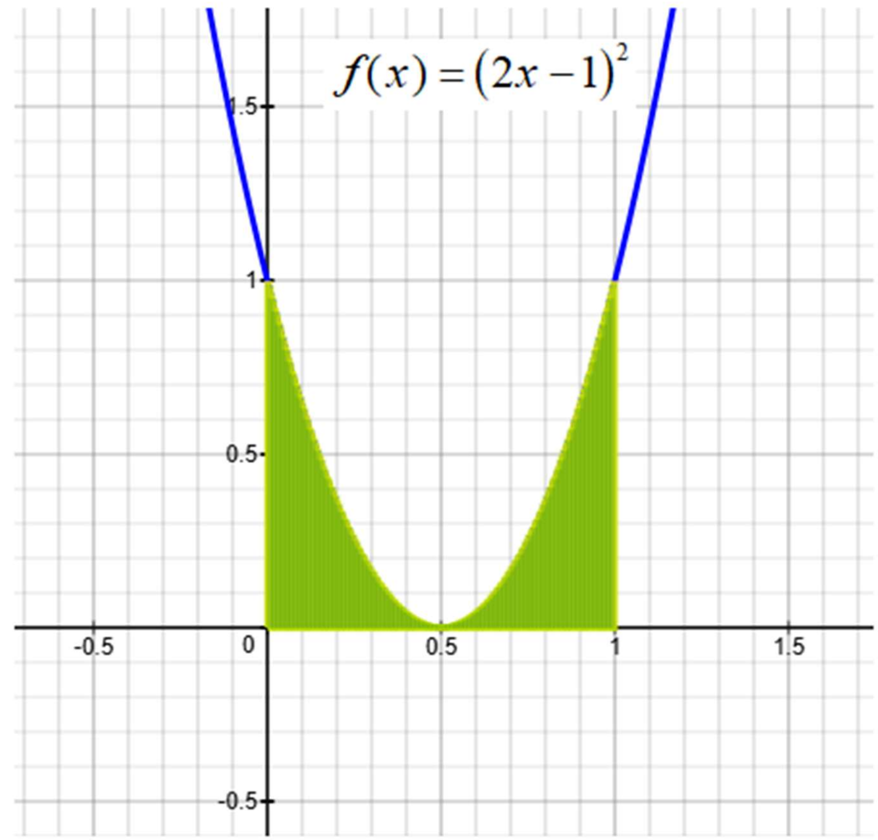
Hint: $(2x-1)^2 = (2x-1)(2x-1) = 4x^2 - 4x + 1$

b) $F(1) = \underline{\quad ? \quad}$

c) $F(0) = \underline{\quad ? \quad}$

d) $\int_0^1 (2x-1)^2 \cdot dx = F(1) - F(0) = \underline{\quad ? \quad}$

e) Area of shaded region $= \underline{\quad ? \quad}$



5) Evaluate the definite integral $\int_1^2 \left(\frac{5}{x^3} - 1\right) \cdot dx$

a) Antiderivate $F(x) = \int \left(x^2 - 1\right) \cdot dx =$ ___?___

Hint: $\frac{5}{x^3} - 1 = 5x^{-3} - 1$

b) $F(2) =$ ___?___

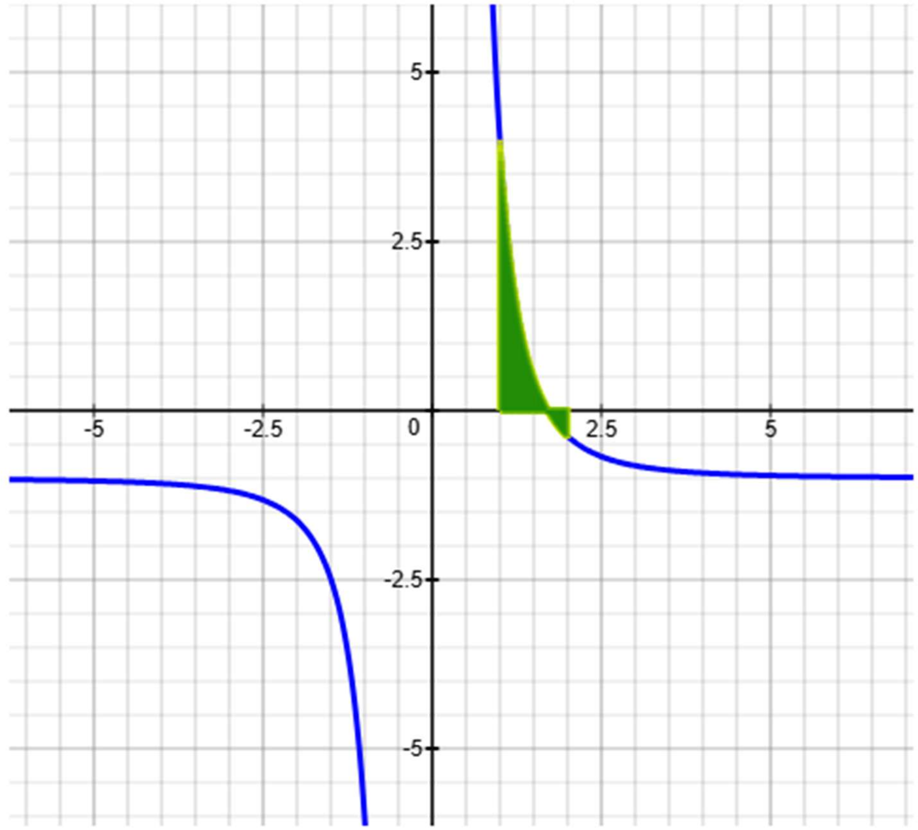
c) $F(1) =$ ___?___

d) $\int_1^2 \left(\frac{5}{x^3} - 1\right) \cdot dx = F(2) - F(1) =$ ___?___

e) Area of shaded region $=$ ___?___

Note: Area above x-axis is positive.

Area below x-axis is negative.



6) Evaluate the definite integral $\int_1^4 \left(\frac{5x-2}{\sqrt{x}} \right) \cdot dx$

a) Antiderivate $F(x) = \int (x^2 - 1) \cdot dx =$ ___?___

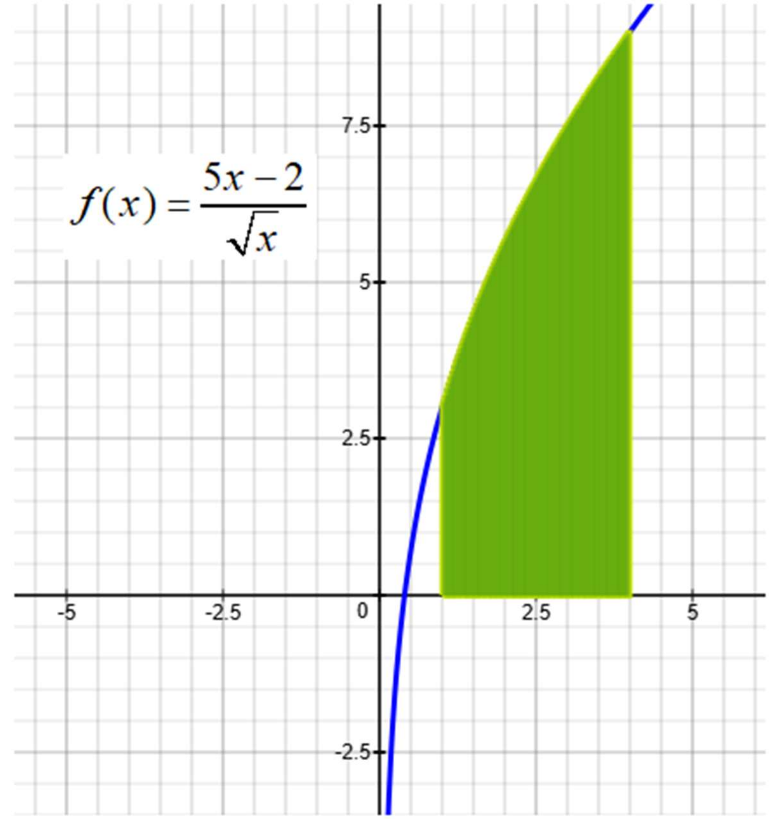
Hint: $\frac{5x-2}{\sqrt{x}} = \frac{5x}{x^{1/2}} - \frac{2}{x^{1/2}} = 5x^{1-1/2} - 2x^{-1/2} = 5x^{1/2} - 2x^{-1/2}$

b) $F(4) =$ ___?___

c) $F(1) =$ ___?___

d) $\int_1^4 \left(\frac{5x-2}{\sqrt{x}} \right) \cdot dx = F(3) - F(1) =$ ___?___

e) Area of shaded region $=$ ___?___



7) Evaluate the definite integral $\int_0^1 \left(\frac{4x-\sqrt{x}}{5}\right)\cdot dx$

a) Antiderivate $F(x)=\int\left(\frac{4x-\sqrt{x}}{5}\right)\cdot dx = \underline{\quad ? \quad}$

Hint: $\frac{4x-\sqrt{x}}{5}=\frac{4x}{5}-\frac{\sqrt{x}}{5}=\frac{4}{5}x-\frac{1}{5}x^{1/2}$

b) $F(1)=\underline{\quad ? \quad}$

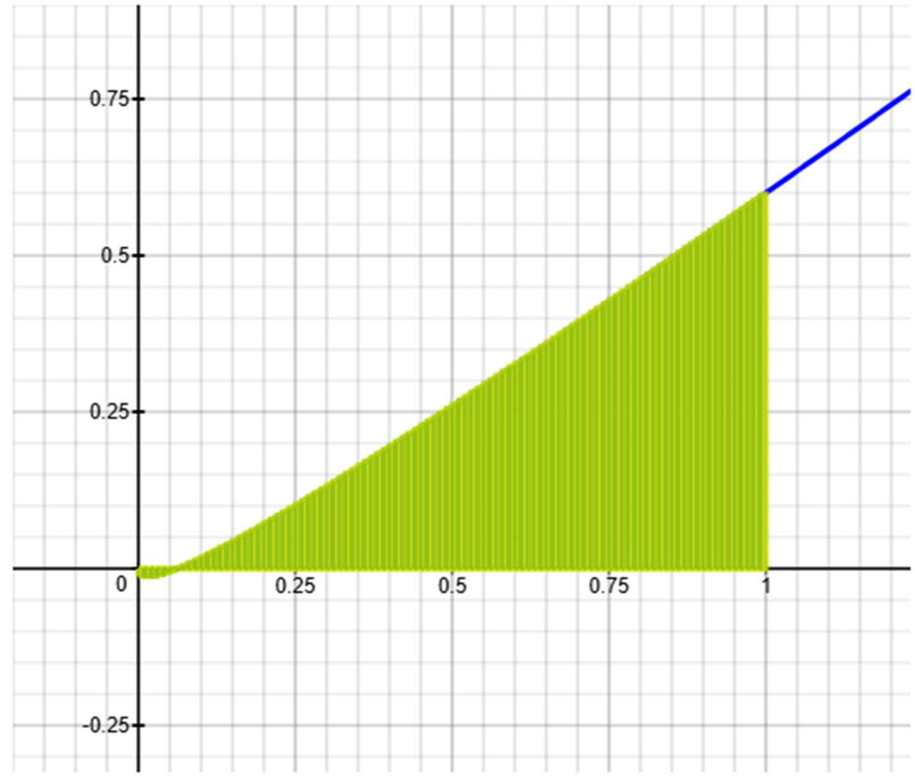
c) $F(0)=\underline{\quad ? \quad}$

d) $\int_0^1\left(\frac{4x-\sqrt{x}}{5}\right)\cdot dx = F(1)-F(0)=\underline{\quad ? \quad}$

e) Area of shaded region $=\underline{\quad ? \quad}$

Note: Area above x-axis is positive.

Area below x-axis is negative.



8) Evaluate the definite integral $\int_0^5 (2x-4)\cdot dx$

a) Antiderivate $F(x) = \int (2x-4)\cdot dx = \underline{\quad ? \quad}$

b) $F(5) = \underline{\quad ? \quad}$

c) $F(0) = \underline{\quad ? \quad}$

d) $\int_0^5 (2x-4)\cdot dx = F(5) - F(0) = \underline{\quad ? \quad}$

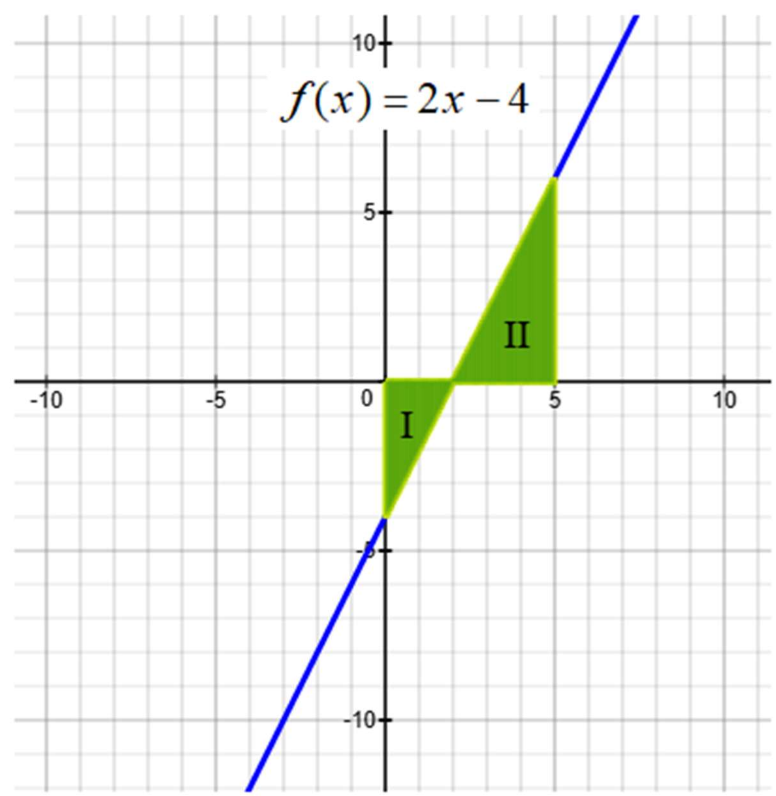
e) Area of Triangle I $= \underline{\quad ? \quad}$

f) Area of Triangle II $= \underline{\quad ? \quad}$

g) $(\text{Area of Triangle II}) - (\text{Area of Triangle I}) = \underline{\quad ? \quad}$

Note: Area above x-axis is positive.

Area below x-axis is negative.



9) Evaluate the definite integral $\int_{-1}^{1}\left(x^{1/3}-2x^{2/3}\right)\cdot dx$

a) Antiderivate $F(x)=\int\left(x^{1/3}-2x^{2/3}\right)\cdot dx =$ ___?___

b) $F(1)=$ ___?___

c) $F(-1)=$ ___?___

d) $\int_{-1}^{1}\left(x^{1/3}-2x^{2/3}\right)\cdot dx = F(1)-F(-1) =$ ___?___

e) Area of shaded region $=$ ___?___

Note: Area above x-axis is positive.

Area below x-axis is negative.